SET

01

Pre-NEET (2025-26)

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 26/04/2026

ANSWER KEY

1.	(3)	46.	(2)	91.	(2)	136.	(1)
2.	(3)	47.	(2)	92.	(2)	137.	(3)
3.	(4)	48.	(3)	93.	(2)	138.	(2)
4.	(4)	49.	(3)	94.	(4)	139.	(3)
5.	(2)	50.	(2)	95.	(3)	140.	(3)
6.	(2)	51.	(2)	96.	(4)	141.	(1)
7.	(4)	52.	(4)	97.	(1)	142.	(2)
8.	(2)	53.	(3)	98.	(1)	143.	(3)
9.	(3)	54.	(1)	99.	(1)	144.	(3)
10.	(2)	55.	(3)	100.	(2)	145.	(2)
11.	(2)	56.	(1)	101.	(3)	146.	(2)
12.	(3)	57.	(1)	102.	(2)	147.	(3)
13.	(3)	58.	(2)	103.	(3)	148.	(3)
14.	(2)	59.	(3)	104.	(2)	149.	(2)
15.	(3)	60.	(2)	105.	(3)	150.	(3)
16.	(1)	61.	(2)	106.	(2)	151.	(3)
17.	(1)	62.	(1)	107.	(1)	152.	(3)
18.	(3)	63.	(4)	108.	(4)	153.	(4)
19.	(2)	64.	(1)	109.	(1)	154.	(4)
20.	(2)	65.	(3)	110.	(3)	155.	(2)
21.	(3)	66.	(4)	111.	(3)	156.	(3)
22.	(1)	67.	(4)	112.	(3)	157.	(4)
23.	(3)	68.	(4)	113.	(3)	158.	(1)
24.	(4)	69.	(3)	114.	(2)	159.	(2)
25.	(1)	70.	(1)	115.	(3)	160.	(3)
26.	(3)	71.	(1)	116.	(4)	161.	(2)
27.	(2)	72.	(2)	117.	(4)	162.	(2)
28.	(4)	73.	(4)	118.	(3)	163.	(3)
29.	(2)	74.	(2)	119.	(1)	164.	(4)
30.	(4)	75.	(3)	120.	(4)	165.	(3)
31.	(1)	76.	(2)	121.	(2)	166.	(3)
32.	(3)	77.	(2)	122.	(2)	167.	(2)
33.	(1)	78.	(3)	123.	(3)	168.	(2)
34.	(3)	79.	(2)	124.	(1)	169.	(2)
35.	(4)	80.	(1)	125.	(2)	170.	(4)
36.	(4)	81.	(2)	126.	(4)	171.	(1)
37.	(4)	82.	(1)	127.	(3)	172.	(4)
38.	(4)	83.	(4)	128.	(2)	173.	(2)
39.	(2)	84.	(1)	129.	(4)	174.	(2)
40.	(2)	85.	(3)	130.	(4)	175.	(2)
41.	(1)	86.	(2)	131.	(1)	176.	(3)
42.	(4)	87.	(3)	132.	(1)	177.	(1)
43.	(2)	88.	(4)	133.	(3)	178.	(4)
44.	(3)	89.	(2)	134.	(3)	179.	(4)
45.	(2)	90.	(3)	135.	(4)	180.	(2)

Hints & Solutions

1. (3)

$W = \frac{nR(T_2 - T_1)}{\gamma - 1} \Rightarrow 22500 = \frac{2(25/3)(T_2 - 300)}{2/3}$

$22500 = 25(T_2 - 300) \Rightarrow T_2 = 1200$ K

Use the adiabatic relation $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$.

$\left(\frac{V_2}{V_1}\right)^{5/3-1} = \frac{T_1}{T_2}$

$\left(\frac{V_2}{V_1}\right)^{2/3} = \frac{300}{1200} = \frac{1}{4}$

$\frac{V_2}{V_1} = \left(\frac{1}{4}\right)^{3/2} = \frac{1}{8}$

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2. (3)

(A) Magnetic Flux (ϕ):

$\phi = V \cdot t = \left(\frac{J}{C}\right) \cdot s = \frac{J}{(C/s)} = J \cdot A^{-1}$ (Matches III)

(B) Self-Inductance (L):

$L = \frac{V}{di/dt} = \left(\frac{V}{A}\right) \cdot s = \Omega \cdot s$ (Matches IV)

(C) Permittivity (ϵ_0):

$\epsilon_0 = \frac{C \cdot d}{A} = \frac{(\text{Charge / Voltage})}{\text{meter}} = C \cdot V^{-1} \cdot m^{-1}$

(Matches I)

(D) Mobility (μ): $\mu = \frac{v_d}{E} = \frac{m/s}{V/m} = m^2 \cdot V^{-1} \cdot s^{-1}$

(Matches II)

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3. (4)

Switch Open: Current $I = 0$, so the voltmeter reads the full emf: $V = 6.0$ V.

Switch Closed: Current $I = \frac{6.0 \text{ V}}{2.0\,\Omega + 3.0\,\Omega} = 1.2$ A.

The voltmeter reads the terminal voltage:

$V = I \times R_{ext} = 1.2 \text{ A} \times 3.0\,\Omega = 3.6$ V.

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4. (4)

A closed organ pipe (closed at one end, open at the other) only produces odd harmonics. The allowed frequencies are $f_1, 3f_1, 5f_1, 7f_1, \ldots$ Fundamental = 1st harmonic $= f_1 = 250$ Hz.

First overtone = 3rd harmonic $= 3f_1$.

Second overtone = 5th harmonic $= 5f_1$.

Frequency $= 5 \times 250 = 1250$ Hz.

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5. (2)

For circular motion, $\frac{mv^2}{r} = \frac{ke^2}{r^2} \Rightarrow mv^2 = \frac{ke^2}{r}$.

Kinetic energy $K = \frac{1}{2}mv^2 = \frac{ke^2}{2r}$.

Potential energy $U = -\frac{ke^2}{r}$. Thus, $|U| = 2K$, and

total energy $E = K + U = -\frac{ke^2}{2r}$ (negative).

Electrostatic force of attraction provides necessary centripetal force.

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6. (2)

The 2 kg block leaves the ground when rope tension becomes

$T = 2g = 20$ N

Since spring is massless,

$T = kx$

$20 = 40x$

$x = 0.5$ m

The 5 kg block moves down by 0.5 m.

Loss in PE

$= 5 \times 10 \times 0.5 = 25$ J

Spring energy

$= \frac{1}{2}kx^2 = 5$ J

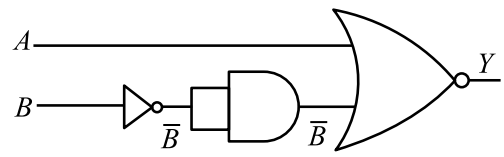
Remaining $KE = 20$ J

$\frac{1}{2}(5)v^2 = 20$

$v = 2\sqrt{2}$ m/s

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7. (4)



$Y = \overline{A + \overline{B}}$

$Y = \overline{A} \cdot B$

8. (2)

Q-value = Final binding energy - Initial binding energy $= E_2N_2 + E_3N_3 - E_1N_1$

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9. (3)

Rotational kinetic energy $K = \frac{L^2}{2I} \Rightarrow L = \sqrt{2IK}$.

Since K is the same, $L \propto \sqrt{I}$.

Moment of inertia for solid sphere $I_S = \frac{2}{5}MR^2$.

Hollow sphere $I_H = \frac{2}{3}MR^2$.

Ratio $\frac{L_S}{L_H} = \sqrt{\frac{I_S}{I_H}} = \sqrt{\frac{2/5}{2/3}} = \sqrt{\frac{3}{5}}$.

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10. (2)

Magnetic susceptibility

$\chi = \mu_r - 1 = 0.9983 - 1 = -0.0017$.

Since $\mu_r < 1$ and χ is slightly negative, the material actively repels magnetic field lines and is therefore firmly classified as Diamagnetic.

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11. (2)

Pressure from piston

$= \frac{mg}{A}$

Equivalent pressure head

$= \frac{mg}{(\rho A)}$

Using Bernoulli principle,

$v = \sqrt{2\left(gh + \frac{mg}{(\rho A)}\right)}$

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12. (3)

The magnetic field produced by the long wire at a distance $r = 0.2$ m is:

$B = \frac{\mu_0 I}{2\pi r} = \frac{4\pi \times 10^{-7} \times 10}{2\pi \times 0.2} = 10^{-5}$ T.

Since the electron is moving parallel to the wire, its velocity vector is perpendicular to the circular magnetic field lines ($\theta = 90^\circ$).

Force $F = qvB\sin(90^\circ)$

$= (1.6 \times 10^{-19}) \times (10^5) \times (10^{-5}) \times 1 = 1.6 \times 10^{-19}$ N

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13. (3)

Equation of SHM from mean position:

$x = A\sin(\omega t)$.

For $x = \frac{A}{2}$:

$\frac{A}{2} = A\sin(\omega t_1) \Rightarrow \sin(\omega t_1) = \frac{1}{2} \Rightarrow \omega t_1 = \frac{\pi}{6}$.

Since $\omega = 2\pi / T$, we have $\left(\frac{2\pi}{T}\right)t_1 = \frac{\pi}{6} \Rightarrow t_1 = \frac{T}{12}$

The total time to reach the extreme position ($x = A$) from the mean position is $\frac{T}{4}$.

So, $t_2 = \left(\frac{T}{4}\right) - t_1 = \frac{T}{4} - \frac{T}{12} = \frac{2T}{12} = \frac{T}{6}$.

Ratio $\frac{t_1}{t_2} = \frac{T/12}{T/6} = \frac{1}{2}$, or 1: 2.

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14. (2)

Fundamental frequency of an open organ pipe is $f = \frac{v}{2L}$.

Beat frequency $\Delta f = f_1 - f_2 = \frac{v}{2L_1} - \frac{v}{2L_2}$.

Given $L_1 = 0.5$ m and $L_2 = 0.505$ m.

$3 = \frac{v}{2(0.5)} - \frac{v}{2(0.505)} = \frac{v}{1.00} - \frac{v}{1.01}$.

$3 = v\left(1 - \frac{1}{1.01}\right) = v\left(\frac{1.01 - 1}{1.01}\right) = v\left(\frac{0.01}{1.01}\right)$.

$v = \frac{3 \times 1.01}{0.01} = 303$ m/s.

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15. (3)

Potential

$V = \frac{kq}{r}$

Equal potential differences occur with increasing spacing outward.

Thus

$t_1 < t_2$

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16. (1)

Filling half the thickness parallel to the plates equates to forming two capacitors in series. The air-filled half has capacitance $C_1 = \frac{\epsilon_0 A}{d/2} = 2C$.

The dielectric-filled half has capacitance

$$C_2 = \frac{K\epsilon_0 A}{d/2} = 2KC = 2(3)C = 6C.$$

Equivalent capacitance in series:

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{2C \times 6C}{2C + 6C} = \frac{12C^2}{8C} = 1.5C.$$

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17. (1)

For an ideal gas, there are no intermolecular attractive forces, so internal energy is purely kinetic and depends only on temperature. Statement I is true. In an isothermal process, $\Delta T = 0$, so $\Delta U = 0$.

By the first law, $\Delta Q = \Delta U + \Delta W \Rightarrow \Delta Q = \Delta W$.

The heat added is entirely converted to work. Statement II is true.

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18. (3)

Surface Gravity: Since M is constant, $g_s = \frac{GM}{R^2}$,

so $g_s \propto \frac{1}{R^2}$.

At depth $d = R/2$, the term $(1 - d/R)$ becomes $(1 - 1/2) = 1/2$ for all planets.

Therefore, $g_d = \left(\frac{GM}{R^2}\right) \times \frac{1}{2}$.

This means $g_d \propto \frac{1}{R^2}$.

Ratio: $g_1 : g_2 : g_3 : g_4 = \frac{1}{1^2} : \frac{1}{2^2} : \frac{1}{3^2} : \frac{1}{4^2}$

$= 1 : \frac{1}{4} : \frac{1}{9} : \frac{1}{16}$

Multiply by 144 (the LCM) to get: 144 : 36 : 16 : 9.

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19. (2)

$\vec{V}_A = 10\cos 53^\circ \hat{i} + 10\sin 53^\circ \hat{j} = 6\hat{i} + 8\hat{j}$

$\vec{V}_B = 4\hat{i}$

$\vec{V}_{com} = \frac{2(6\hat{i} + 8\hat{j}) + 2(4\hat{i})}{4}$

$= \frac{12\hat{i} + 16\hat{j} + 8\hat{i}}{4} = \frac{20\hat{i} + 16\hat{j}}{4} = 5\hat{i} + 4\hat{j}$

$|\vec{V}_{com}| = \sqrt{25 + 16} = \sqrt{41}$ m/s.

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20. (2)

When the source is shifted upwards, the light waves have to travel a slightly shorter path to reach the upper slit and a longer path to reach the lower slit. To maintain a path difference of exactly zero (the condition for the central maximum), the observation point on the screen must shift geometrically in the opposite direction (downwards) to compensate for this initial path difference.

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21. (3)

The only torque acting on the rod about the hinge immediately upon release is due to gravity.

The weight Mg acts exactly through the center of mass, which is at a distance of $\frac{L}{2}$ from the hinge.

Torque $\tau = Mg\left(\frac{L}{2}\right)$.

Moment of inertia of a rod about its end is $I = \frac{ML^2}{3}$

Using $\tau = I\alpha \Rightarrow Mg\left(\frac{L}{2}\right) = \left(\frac{ML^2}{3}\right)\alpha$.

$\alpha = \frac{3g}{2L}$.

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22. (1)

An electron (negatively charged) naturally experiences a force opposite to the electric field, which means it is accelerated from the *p*-side towards the *n*-side.

The potential difference is 0.7 V,

$\Delta U = q\Delta V = (-e)(-0.7\text{ V}) = +0.7$ eV in kinetic terms.

Final Kinetic Energy = Initial Kinetic Energy + Work done by the electric field

$= 0.5\text{ eV} + 0.7\text{ eV} = 1.2\text{ eV}$.

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23. (3)

Voltage across galvanometer

$V = I_g G = 0.05 \times 50 = 2.5$ V

Shunt current

$= 5 - 0.05 = 4.95$ A

Shunt resistance

$S = \frac{2.5}{4.95} \approx 0.505\ \Omega$

Using $S = \frac{\rho L}{A}$

$L \approx 3$ m

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24. (4)

Since the elevator is moving upward with a constant velocity, its acceleration is zero. This means the net force is zero, and the upward force exerted by the motor must perfectly balance all downward forces.

Downward forces = Weight of elevator (mg) + Frictional force (f)

$F_{\text{motor}} = (1000 \times 10) + 4000$

$F_{motor} = 10000 + 4000 = 14000$ N

The power delivered by a constant force moving an object at a constant velocity is given by $P = F \times v$.

$P = 14000\ \text{N} \times 2\ \text{m/s} = 28000\ \text{W} = 28\ \text{kW}$.

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25. (1)

Assertion A is true. The tangent of the angle of friction θ equals the coefficient of static friction μ_s.

The tangent of the angle of repose α is also equal to μ_s. Thus, $\theta = \alpha$.

Reason R is false. The maximum static frictional force depends on normal reaction ($f_{s,max} = \mu_s N$) and coefficient of friction.

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26. (3)

Therefore, the nuclear density (Mass/Volume) is totally independent of the mass number and remains virtually constant for all stable nuclei. The ratio is exactly 1: 1.

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27. (2)

From Einstein's photoelectric equation:

$K_{max} = h\nu - \phi$.

Case 1: $K = h\nu - \phi$

Case 2: $3K = h(2\nu) - \phi$

Dividing the equations: $\frac{1}{3} = \frac{h\nu - \phi}{2h\nu - \phi}$

$\Rightarrow 2h\nu - \phi = 3h\nu - 3\phi \Rightarrow 2\phi = h\nu \Rightarrow \phi = \frac{h\nu}{2}$.

Since $\phi = h\nu_0$, the threshold frequency $\nu_0 = \frac{\nu}{2}$.

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28. (4)

$\lambda = \frac{h}{\sqrt{2meV_0}}$

$\lambda^4 = \left(\frac{h^4}{4m^2e^2}\right) \cdot \frac{1}{V_0^2} \Rightarrow \lambda^4 \propto \frac{1}{V_0^2}$

The equation is of the form $y = mx$, where $y = \lambda^4$ and $x = \frac{1}{V_0^2}$. This represents a straight line.

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29. (2)

Magnitude of induced emf is $|e| = M\frac{|\Delta I|}{\Delta t}$.

Rearrange for ΔI : $|\Delta I| = \frac{|e| \cdot \Delta t}{M}$.

$|\Delta I| = \frac{0.6 \times 0.02}{1.5 \times 10^{-3}}$

$|\Delta I| = \frac{0.012}{0.0015} = \frac{120}{15} = 8$ A

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30. (4)

Orbital velocity $v_o = \sqrt{\frac{GM}{r}}$, where $r = R + h$.

Escape velocity from Earth's surface $v_e = \sqrt{\frac{2GM}{R}}$.

Given $v_o = \frac{v_e}{2} \Rightarrow \sqrt{\frac{GM}{R+h}} = \frac{1}{2}\sqrt{\frac{2GM}{R}} = \sqrt{\frac{GM}{2R}}$.

Squaring both sides gives

$\frac{1}{R+h} = \frac{1}{2R} \Rightarrow R + h = 2R \Rightarrow h = R$.

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31. (1)

The rms speed of gas molecules is $v_{rms}=\sqrt{\frac{3RT}{M}}$.

For v_{rms} to be strictly equal: $\frac{T_H}{M_H}=\frac{T_O}{M_O}$.

Molecular mass of Hydrogen $M_H=2$ g/mol.

Molecular mass of Oxygen $M_O=32$ g/mol.

$\frac{T_H}{2}=\frac{320}{32}\Rightarrow\frac{T_H}{2}=10\Rightarrow T_H=20$ K.

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32. (3)

Efficiency $\eta=\frac{V_sI_s}{V_pI_p}\times100\%$

A: $\frac{100\times9}{200\times5}=\frac{900}{1000}=90\%$

B: $\frac{22\times85}{220\times10}=\frac{1870}{2200}=85\%$

C: $\frac{50\times3}{100\times2}=\frac{150}{200}=75\%$ (Minimum)

D: $\frac{100\times19}{500\times4}=\frac{1900}{2000}=95\%$

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33. (1)

$F=m\times a$

$m=\frac{F}{a}$

$[M]=\frac{[F]}{[A]}=[F^1A^{-1}]$

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34. (3)

$I_{vac}=\frac{1}{2}\epsilon_0cE_{vac}^2$

In the medium, $v=\frac{c}{n}$.

Since it is a non-magnetic medium, the permittivity is $\epsilon=\epsilon_0\epsilon_r$. We also know that $n=\sqrt{\epsilon_r\mu_r}$.

Since $\mu_r=1$, then $n^2=\epsilon_r$. So, $\epsilon=n^2\epsilon_0$.

$I_{med}=\frac{1}{2}(n^2\epsilon_0)\left(\frac{c}{n}\right)E_{med}^2=\frac{1}{2}n\epsilon_0cE_{med}^2$

$\frac{1}{2}\epsilon_0cE_{vac}^2=\frac{1}{2}n\epsilon_0cE_{med}^2$

$E_{vac}^2=nE_{med}^2$

$\frac{E_{vac}}{E_{med}}=\sqrt{n}$

$\frac{E_{vac}}{E_{med}}=\sqrt{4}=2$

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35. (4)

Image is inverted, so it is real. u and v both are negative. Magnification is $\frac{1}{3}$, therefore, $v=\frac{u}{3}$

Given, $u=-30$ cm, $v=-10$ cm

Using the mirror formula,

$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

We have, $\frac{1}{f}=\frac{1}{-10}+\frac{1}{-30}=\frac{-2}{15}$

$f=-7.5$ cm

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36. (4)

$\phi=\vec{B}\cdot\vec{A}$

$=(\mu H)A$

$\mu=\frac{\phi}{HA}=\frac{4\times10^{-5}}{500\times0.5\times10^{-4}}$

$\mu=1.6\times10^{-3}$ H/m

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37. (4)

Since $V_L=20$ V and $V_C=20$ V, we have $V_L=V_C$

This indicates the circuit is in a state of electrical resonance.

In a resonant series LCR circuit, the net reactance is zero ($X_L=X_C$), and the impedance is purely resistive ($Z=R$). Thus, the applied voltage and current are perfectly in phase, meaning the phase difference is 0°.

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38. (4)

$\phi_{net}=\frac{q_{enclosed}}{\epsilon_0}$

$=\frac{\frac{3q}{8}-\frac{3q}{8}+\frac{6q}{8}-\frac{6q}{8}}{\epsilon_0}$

$=0$

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39. (2)

Area under $a-t$ graph = change in velocity

$v_{max}-0=\frac{1}{2}\times6\times5$

$v_{max}=15$ m/s

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40. (2)

Velocity $\vec{v} = u\cos\theta\hat{i} + (u\sin\theta - gt)\hat{j}$.

Position $\vec{r} = (u\cos\theta)t\hat{i} + (u\sin\theta t - \frac{1}{2}gt^2)\hat{j}$.

For perpendicular vectors, $\vec{v}\cdot\vec{r} = 0$.

$(u^2\cos^2\theta)t + (u\sin\theta - gt)(u\sin\theta t - \frac{1}{2}gt^2) = 0$.

$u^2\cos^2\theta + u^2\sin^2\theta - \frac{1}{2}ugt\sin\theta$

$$-ugt\sin\theta + \frac{1}{2}g^2t^2 = 0$$

$\frac{1}{2}g^2t^2 - \frac{3}{2}ugt\sin\theta + u^2 = 0$

$\Rightarrow g^2t^2 - 3ugt\sin\theta + 2u^2 = 0$

For a real time t, the discriminant $\Delta \geq 0$.

$(3ug\sin\theta)^2 - 4(g^2)(2u^2) \geq 0$

$\Rightarrow 9u^2g^2\sin^2\theta \geq 8u^2g^2$

$\Rightarrow \sin^2\theta \geq \frac{8}{9} \Rightarrow \sin\theta \geq \frac{2\sqrt{2}}{3}$

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41. (1)

$\Delta V = \gamma V\Delta T$

$\Delta V = 3\alpha V\Delta T = 3\times 23\times 10^{-6}\times\frac{4}{3}\pi\times(0.1)^3\times 100$

$28.9\times 10^{-6}\ \text{m}^3 = 28.9\ \text{cc}$

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42. (4)

$\tau = \vec{M}\times\vec{B}$

$\tau = NIBA\sin 90^\circ$

$= 300\times 10^{-4}\times(3\times 10^{-2}\times 10\times 10^{-2})\times 0.2$

$\tau = 1.8\times 10^{-5}\ \text{Nm}$

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43. (2)

The square has 4 sides, each having a resistance of $10\,\Omega$. Connecting across one side puts one $10\,\Omega$ resistor in parallel with the series combination of the other three sides ($30\,\Omega$).

Equivalent resistance of the square

$R_p = \frac{10\times 30}{10+30} = 7.5\,\Omega$.

Total resistance of the circuit

$R_{eq} = 7.5 + 2.5 = 10\,\Omega$

Total current from battery $I = \frac{20}{10} = 2$ A

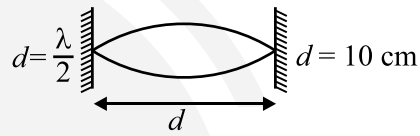
Voltage drop across the parallel combination

$V = I\times R_p = 2\times 7.5 = 15$ V.

Power dissipated in the $30\,\Omega$ arm (the other three sides) = $\frac{V^2}{R} = \frac{15^2}{30} = \frac{225}{30} = 7.5$ W.

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44. (3)



$\lambda = \frac{2\pi}{k} = \frac{2\pi}{0.314} = 20$ cm

Smallest length $d = \frac{\lambda}{2} = 10$ cm

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45. (2)

Initial momentum is zero. By conservation of momentum, the vector sum of final momenta must be zero.

Momenta of the first two mutually perpendicular fragments: $p_1 = 3\times 4 = 12$ kg m/s and $p_2 = 3\times 4 = 12$ kg m/s.

Their resultant momentum

$p_{12} = \sqrt{12^2 + 12^2} = 12\sqrt{2}$ kg m/s.

To conserve momentum, the third fragment must have an equal and opposite momentum:

$p_3 = 12\sqrt{2}$ kg m/s.

Given velocity $v_3 = 6\sqrt{2}$ m/s, so its mass

$m_3 = \frac{p_3}{v_3} = \frac{12\sqrt{2}}{6\sqrt{2}} = 2$ kg.

Total mass $= m_1 + m_2 + m_3 = 3 + 3 + 2 = 8$ kg.

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[Contd...

46. (2)

Insulin is released in response to the rapid rise in blood glucose level. Abnormally low level of thyroxine leads to hypothyroidism which is characterised by lethargyness and obesity.

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47. (2)

$$\text{Mass of nitrogen} = \frac{14}{17} \times 0.17\ \text{g}$$

$$\%\ \text{of N} = \frac{\frac{14}{17} \times 0.17}{2} \times 100 = 7$$

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48. (3)

		Unpaired electron
$[Ni(CO)_4]$	$3d^{10}, sp^3$	0
$[NiCl_4]^{2-}$	$3d^8, sp^3$	2
$[Co(NH_3)_4Cl_2]Cl$	$3d^6, d^2sp^3$	0
$Na_3[CoF_6]$	$3d^6, sp^3d^2$	4
Na_2O_2	O_2^{2-}	0
CsO_2	O_2^{-1}	1

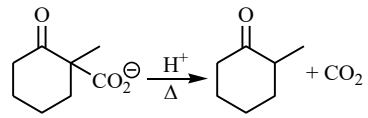
(NCERT Ed. 2025-26, 11th, Page No. 325)

49. (3)

Paramagnetic compounds are 4 $[NiCl_4]^{2-}$, O_2, $Na_3[CoF_6]$, CsO_2

(NCERT Ed. 2025-26, 12th, Page No. 139)

50. (2)



(salt of β-ketoacid)
(A)

(NCERT Ed. 2025-26, 12th, Page No. 104)

51. (2)

Anode:

$$2Cl^- \rightarrow \underset{\substack{1\ \text{mol} \\ (22.4\ \text{L at STP})}}{Cl_2} + \underset{2F}{2e^-}$$

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52. (4)

- 0.3 g of $Ca(OH)_2$ in 500 mL solution

  Molar mass of $Ca(OH)_2$

  $40 + 2(16 + 1) = 74$

  Moles: $n = \frac{0.3}{74} = 0.00405$

  Volume = 500 mL = 0.5 L

  Molarity of $Ca(OH)_2$: $\frac{0.00405}{0.5} = 0.00810$

  Each mole gives 2 moles OH^-, so

  $[OH^-] = 2 \times 0.00810 = 0.0162$

  $pOH = -\log(0.0162) \approx 1.79$

  $pH = 14 - 1.79 = 12.21$

- 1 mL of 13.6 M HCl diluted to 1 L

  Use dilution: $M_1V_1 = M_2V_2$

  $13.6 \times 1\ \text{mL} = M_2 \times 1000\ \text{mL}$

  $M_2 = \frac{13.6}{1000} = 0.0136$

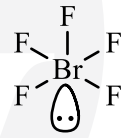
  Since HCl is strong: $[H^+] = 0.0136$

  $pH = -\log(0.0136) \approx 1.87$

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53. (3)

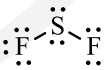
- BrF_5 :



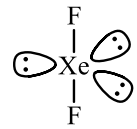
- PF_5 :



- SF_2 :



- XeF_2 :



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54. (1)

- On increasing the concentration of solute, vapour pressure of solution decreases.
- Vapour pressure is not a colligative property.

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55. (3)

$$2C(s) + 2H_2 \rightarrow C_2H_4;\quad \Delta_rH = \Sigma(BE)_R - \Sigma(BE)_P$$

$$= (2 \times 710 + 2 \times 436) - (4 \times 414 + 611) = 25\ \text{kJ mol}^{-1}$$

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[Contd...

56. (1)

For acidic buffer solution:

$$pH = pK_a + \log_{10}\frac{[Salt]}{[Acid]}$$

$$5 = 6 + \log\left[\frac{Salt}{Acid}\right]; \quad -1 = \log\left[\frac{Salt}{Acid}\right]$$

$$\frac{[Salt]}{[Acid]} = 10^{-1} = 0.1$$

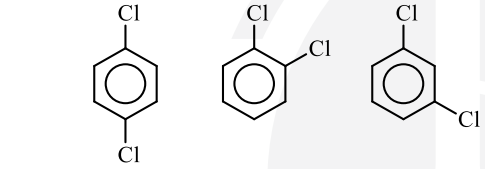
(NCERT Ed. 2025-26, 11th, Page No. 203)

57. (1)

- On addition of starch after the liberation of iodine from the reaction of Cu^{2+} ions on iodide ions, an intense blue colour appears.
- $KMnO_4$ acts as the self indicator.

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58. (2)

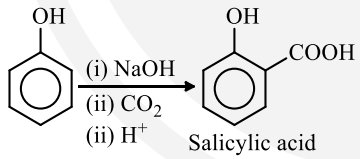


mp/K	323	256	251

(NCERT Ed. 2025-26, 12th, Page No. 168)

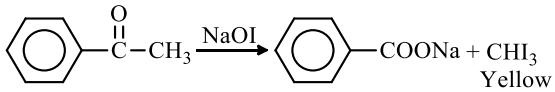
59. (3)

Kolbe's reaction:



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60. (2)



(NCERT Ed. 2025-26, 12th, Page No. 240)

61. (2)

$$Cu^{2+} + S^{2-} \rightarrow \underset{ppt}{CuS}; \quad Cu^{2+} + SO_4^{2-} \rightarrow \underset{(soluble)}{CuSO_4}$$

62. (1)

–SH > –OH (acidic strength) because of bigger sizer of sulphur.

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63. (4)

$$K_2 = \frac{1}{t}\ln\frac{100}{4} \quad \ldots (i)$$

$$K_1 = \frac{1}{t}\ln\frac{100}{50} \quad \ldots (ii)$$

$$\frac{K_2}{K_1} = \frac{\ln 25}{\ln 2} = 4.65$$

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64. (1)

$$N_2(g) + 5H^+(aq) + 4e^- \rightarrow N_2H_5^+(aq), \quad E^o = -0.23\,V \quad \ldots (1)$$

$$BiO^+(aq) + 2H^+(aq) + 3e^- \rightarrow Bi(S) + H_2O(\ell), \quad E^o = 0.32\,V \quad \ldots (2)$$

[Reaction (1)] × –3 + [Reaction (2)] × 4, we get

$$4BiO^+(aq) + 3N_2H_5^+(aq) \rightarrow 4Bi(s) + 3N_2(g) + 4H_2O(\ell) + 7H^+(aq)$$

$$\Delta G^o = -3(-4FE^o) + 4(-3FE^o)$$

$$-12FE^o = 12F(-0.23) - 12F(0.32)$$

$$E^o = 0.23 + 0.32 = 0.55$$

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65. (3)

- The metallic radius of Al is more than Ga.
- Density of Al is more than Ga.

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66. (4)

- Due to absence of acidic H-atom, tertiary amines do not react with benzenesulphonyl chloride.
- Aniline does not give Friedel-crafts reaction.

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67. (4)

Proline and arginine are non-essential amino acids.

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68. (4)

- $O \Rightarrow -141$
- $S \Rightarrow -200$
- $Cl \Rightarrow -349$
- $Te \Rightarrow -190$

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[Contd...

69. (3)

Use de Broglie relation: $\lambda = \frac{h}{mv}$

Given:

- $h = 6.626 \times 10^{-34}$ Js
- $m = 0.1$ kg
- $v = 4.37 \times 10^{5}$ ms^{-1}

$$\lambda = \frac{6.626 \times 10^{-34}}{0.1 \times 4.37 \times 10^{5}}$$

$$\lambda = \frac{6.626 \times 10^{-34}}{4.37 \times 10^{4}}$$

$$\lambda \approx 1.52 \times 10^{-38} \text{ m}$$

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70. (1)

0.478 mL = 0.478 g of water; 10^9 g water contain 0.10 g $CHCl_3$

$\therefore$ 0.478 g water contain

$\frac{0.1}{10^9} \times 0.478$ g $CHCl_3$ $\quad \therefore \ n_{CHCl_3} = \frac{0.1}{10^9} \times \frac{0.478}{119.5}$

$\therefore$ No. of molecules $= \frac{0.1}{10^9} \times \frac{0.478}{119.5} \times N_A = 4 \times 10^{-13} \times N_A$

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71. (1)

We use:

Fraction of molecules $= e^{-E_a/RT}$

Given:

- $E_a = 209.5 \times 10^3$ J mol^{-1}
- $R = 8.314$ J mol^{-1}K^{-1}
- $T = 581$ K

$$\frac{E_a}{RT} = \frac{209.5 \times 10^3}{8.314 \times 581} \approx 43.4$$

So, $e^{-E_a/RT} = e^{-43.4} \approx 1.5 \times 10^{-19}$

This is an extremely small number.

- Statement I is correct because the fraction is indeed very small.
- Statement II is correct because a large $\frac{E_a}{RT}$, makes $e^{-E_a/RT}$ extremely small.

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72. (2)

$\Rightarrow$ $Fe(CO)_5$: Total number of valence electrons is 18
: low spin complex.

$\Rightarrow$ Ni(CO)4 : Total number of valence electrons is 18
: low spin complex

$\Rightarrow$ Metal-carbonyl bond strengthens when the oxidation state of metal is lowered.

$\Rightarrow$ The carbonyl C–O bond is stronger in case of increased oxidation state of metal.

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73. (4)

$P_{total} = P_{HNO_3} + P_{NO_2} + P_{H_2O} + P_{O_2}$

$\because \ P_{NO_2} = 4P_{O_2}$ and $P_{H_2O} = 2P_{O_2}$

$\therefore \ P_{total} = P_{HNO_3} + 7P_{O_2}$

$\Rightarrow 30 - 2 = P_{O_2} \times 7 \quad \Rightarrow P_{O_2} = \frac{28}{7} = 4$

$$K_p = \frac{P_{NO_2}^4 \cdot P_{H_2O} \cdot P_{O_2}}{P_{HNO_3}^4} = \frac{(4 \times 4)^4 \times (2 \times 4)^2 \times 4}{2^4} = 2^{20}$$

$$K_p = K_c(RT)^{\Delta n_g} = K_c(0.08 \times 400)^3 \Rightarrow K_c = \frac{2^{20}}{(32)^3} = 32$$

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74. (2)

The redox reaction is:

$$Cr_2O_7^{2-} + 3SO_2 + 2H^+ \rightarrow 2Cr^{3+} + 3SO_4^{2-} + H_2O$$

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75. (3)

The given orbital represent a π-molecular orbital.

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76. (2)

Using: $\alpha = \sqrt{\frac{K_a}{C}}$

Here, $C \approx 0.3265$

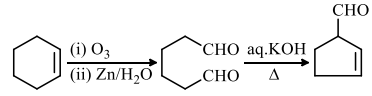
So, $\alpha = \sqrt{\frac{1.4 \times 10^{-3}}{0.3265}} \approx 0.0655$

For a weak monobasic acid: $i = 1 + \alpha$

$i = 1 + 0.0655 = 1.0655 \approx 1.07$

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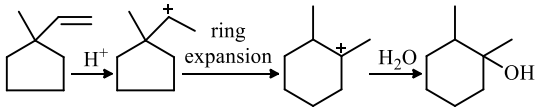
77. (2)



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[Contd...

78. (3)



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79. (2)


is most stable as it is aromatic and has more number of resonating structures.

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80. (1)

- The tanks used by scuba divers are filled with air diluted with 11.7% He.
- An azeotropic mixture of ethanol water mixture contains approximately 95% by volume of ethanol.

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81. (2)

- SeO_3^{2-} : trigonal pyramidal
- CH_3^- : trigonal pyramidal
- BH_3: trigonal planar
- CH_3^+ : trigonal planar

Different from NH_3: III and IV only

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82. (1)

- Electron gain enthalpy = –3 eV/atom
  $\Rightarrow -3\times 23 = -69\,\text{kcal/mol}$
- Moles of $A = \frac{10}{30} = \frac{1}{3}\,\text{mol}$
- Total energy released $= \frac{1}{3}\times 69 = 23\,\text{kcal}$

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83. (4)

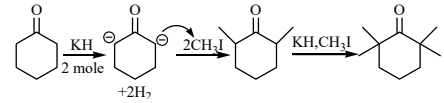
Boiling point: He < Ar < Kr < Xe

(NCERT Ed. 2022-23, 12th, Page No. 208)

84. (1)

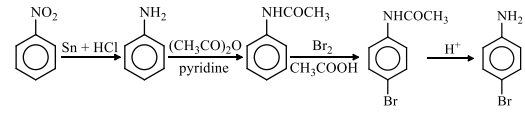
- Passing H_2S in presence of dil. HCl : Cu^{2+}
- NH_4OH in presence of NH_4Cl : Al^{3+}
- $(NH_4)_2CO_3$ in presence of NH_4OH : Ba^{2+}
- Passing H_2S in presence of NH_4OH : Mn^{2+}

85. (3)



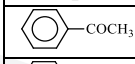
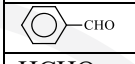
(NCERT Ed. 2025-26, 12th, Page No. 210)

86. (2)



(NCERT Ed. 2025-26, 12th, Page No. 272)

87. (3)

Compound	Fehling test	Tollen's test
	×	×
	×	√
HCHO	√	√
CH_3–COOH	×	×

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88. (4)

- The highest fluoride of Mn is MnF_4.
- The actinoid contraction is due to poor shielding by 5 F electrons

(NCERT Ed. 2025-26, 12th, Page No. 112)

89. (2)

Electronegativity:

F > O > N ≃ Cl > S > P > Be > Mg > Na.

(NCERT Ed. 2025-26, 11th, Page No. 91)

90. (3)

- Percentage of Oxygen: 100 – (69.77 + 11.63) = 18.6%.
- Atomic Ratios:
  0:69.77/12 = 5.81
  H: 11.63/1 = 11.63
  0:18.6/16 = 1.16
- Simplest Ratio: $C_5H_{10}O_1$.
- Calculation: The empirical formula mass (5 × 12 + 10 × 1 + 16) equals 86, which matches the given molecular mass. Thus, the molecular formula is $C_5H_{10}O$.

Forms addition compound with $NaHSO_3$: Indicates a carbonyl group (aldehyde or ketone).

Does not reduce Tollens' reagent: Confirms it is not an aldehyde, but a ketone.

If the compound must strictly not give a positive iodoform test,In that case, is Pentan-3-one

(NCERT Ed. 2025-26, 12th, Page No. 256)

[Contd...

91. (2)

Proteases break down proteins and nucleases break down nucleic acids. If they were freely active in blood, they would digest essential plasma proteins and nucleic acids, disrupt normal physiological functions and cause serious damage.

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92. (2)

In pteridophytes, the main plant body is a sporophyte (which is differentiated into true roots, stems, and leaves), but the prothallus is the gametophyte phase. The prothallus is a small, inconspicuous, multicellular, free-living, mostly photosynthetic thalloid structure. The second stage in life cycle of bryophytes is the leafy stage, which develops from the secondary protonema as a lateral bud. They consist of upright, slender axes bearing spirally arranged leaves. They are attached to the soil through multicellular and branched rhizoids. This stage bears the sex organs.

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93. (2)

The C_3 and C_4 plants respond differently to CO_2 concentrations. The C_4 plants show saturation at about 360 $\mu l L^{-1}$ while C_3 responds to increased CO_2 concentration and saturation is seen only beyond 450 $\mu l L^{-1}$. Thus, current availability of CO_2 levels is limiting to the C_3 plants. The fact that C_3 plants respond to higher CO_2 concentration by showing increased rates of photosynthesis leading to higher productivity has been used for some greenhouse crops.

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94. (4)

Recombination (crossing over) occurs during Prophase I of Meiosis. Because male honey bees are haploid, they do not possess homologous pairs of chromosomes. Consequently, they cannot undergo meiosis or recombination. Honey bees are insects (animals), and during cytokinesis in animal cells, a cleavage furrow is formed, not a cell plate. A cell plate is characteristic of plant cell division. Therefore, the formation of a cell plate cannot take place in the cell cycle of a honey bee.

(NCERT Ed. 2025-26, 11th, Page No. 122)

95. (3)

Plasmodium enters the human body as sporozoites (the infectious form) specifically through the bite of an infected female Anopheles mosquito.

(NCERT Ed. 2025-26, 12th, Page No. 131)

96. (4)

A single U-shaped bone called the hyoid is present at the base of the buccal cavity. The mandible is the jawbone, sphenoid is a cranial bone, and occipital condyles articulate with the atlas.

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97. (1)

Ecologists have discovered that the value of Z (regression coefficient) lies in the range of 0.1 to 0.2, regardless of the taxonomic group or the region (whether it is the plants in Britain, birds in California or molluscs in New York state, the slopes of the regression line are amazingly similar). But, if you analyse the species-area relationships among very large areas like the entire continents, you will find that the slope of the line to be much steeper (Z values in the range 0.6 to 1.2). For example, for frugivorous (fruit-eating) birds and mammals in the tropical forests of different continents, the slope is found to be 1.15.

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98. (1)

Looking at the image, the sequence of nitrogenous bases from the 5' end to the 3' end is: A - T - G - C. The bases in two strands are paired through hydrogen bond (H-bonds) forming base pairs (bp). Adenine forms two hydrogen bonds with Thymine from opposite strand and vice-versa. Similarly, Guanine is bonded with Cytosine with three H-bonds. Therefore, in this fragment, a total of 10 hydrogen bonds will be formed.

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99. (1)

Width of chloroplast - 2 to 4 µm
Length of a mitochondrion - 1.0 to 4.1 µm
Length of mycoplasma - 0.3 µm

(NCERT Ed. 2025-26, 11th, Page No. 100)

100. (2)

When karyokinesis is not followed by cytokinesis it results in a multinucleate condition arises leading to the formation of syncytium. Aneuploidy refers to an abnormal number of chromosomes (e.g., 45 or 47 instead of 46) resulting from errors in chromosome segregation during anaphase. It is not caused by the failure of cytoplasmic division. While abnormal, a binucleated cell does not necessarily die immediately; many organisms (such as tapetal cells of microsporangium wall) are naturally binucleated and function normally. The failure of cytokinesis does not typically cause the cell to automatically "restart" the cell cycle; rather, the cell usually enters a state of polyploidy or remains arrested in that binucleated condition.

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[Contd...

101. (3)

Sweet potato and potato perform the same function (food storage) despite having different anatomical origins, which represents convergent evolution (analogy). Sweet potato is a root modification, whereas potato is a stem modification.

(NCERT Ed. 2025-26, 12th, Page No. 115)

102. (2)

Concanavalin A is specifically categorized as a lectin. Vinblastin is a drug, anthocyanin is a pigment, and ricin is a toxin.

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103. (3)

Impulse transmission across an electrical synapse is always faster (not slower) than across a chemical synapse, as current flows directly from one neuron to the next without waiting for neurotransmitter diffusion.

In a chemical synapse, pre- and post-synaptic membranes are separated by a fluid filled cavity, synaptic cleft.

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104. (2)

Mendelian disorders are mainly determined by alteration or mutation in the single gene (without change in chromosome number). Most common and prevalent Mendelian disorders are Haemophilia, Cystic fibrosis, Sickle-cell anaemia, Colour blindness, Phenylketonuria, Thalassemia, etc. Down's syndrome, Turner's syndrome, Klinefelter's syndrome are common examples of chromosomal disorders. The total number of chromosomes in a normal human cell is 46 (23 pairs). Out of these 22 pairs are autosomes and one pair of chromosomes are sex chromosome. Sometimes, though rarely, either an additional copy of a chromosome may be included in an individual or an individual may lack one of any one pair of chromosomes. These situations are known as trisomy or monosomy of a chromosome, respectively. Such a situation leads to very serious consequences in the individual. Down's syndrome, Turner's syndrome, Klinefelter's syndrome are common examples of chromosomal disorders.

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105. (3)

Genetically modified (GM) crops can be engineered for better nutrient uptake and use efficiency, e.g., crops with enhanced phosphorus or nitrogen utilization. This leads to higher growth and yield with less fertilizer.

Improved mineral use efficiency reduces excessive fertilizer application, which prevents early depletion of soil fertility and maintains soil health.

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106. (2)

A primary spermatocyte completes the first meiotic division (reduction division) leading to the formation of two equal, haploid cells called secondary spermatocytes, which have only 23 chromosomes each.

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107. (1)

Tegmina are hardened forewings from the mesothorax.

Hind wings are membranous, used for flight, from metathorax.

7th-9th sterna form the ovipositor or brood pouch (ootheca) in females.

Anal styles are male-specific structures.

Malpighian tubules are located at the junction of the midgut and hindgut, not the foregut and midgut (where hepatic caeca are found).

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108. (4)

Radial symmetry means the body can be divided into equal halves along multiple planes through a central axis.

Echinodermata (adult starfish) show pentaradial symmetry, Cnidaria have true radial symmetry, and Ctenophora show biradial symmetry (a modified radial form), so all of them possess radial-type symmetry.

Cyclostomata, being vertebrates like lampreys, exhibit bilateral symmetry, where only one plane divides the body into equal halves. Hence, radial symmetry is not present in them.

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109. (1)

Descending limb is highly permeable to water, not electrolytes. Ascending limb is impermeable to water, transports electrolytes. PCT is lined by simple cuboidal epithelium with microvilli, not squamous. DCT secretes H^+ and K^+ to maintain blood pH. Collecting duct is partially permeable to urea, aiding medullary osmolarity.

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110. (3)

Except for hepatitis-B, genital herpes, and HIV infections, all other STIs (including syphilis, chlamydiasis, and trichomoniasis) are completely curable if detected early and treated properly.

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111. (3)

List-I	List-II
Testes	Yellowish and ovoid in shape
Ureters	Act as urinogenital ducts
Sinus venosus	Triangular structure
Conus arteriosus	Present on the ventral side of heart

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112. (3)

Recombinant DNA is formed in prophase I of meiosis I when homologous chromosomes pair up and exchange segments of genetic material between non-sister chromatids. This process is called crossing over, and it generates new combinations of alleles, leading to genetic variation in gametes.

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113. (3)

Basophils are the least abundant WBCs (0.5-1%) and they secrete histamine, serotonin, and heparin involved in inflammatory reactions, not antibodies. B-lymphocytes produce antibodies.

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114. (2)

In the tissues, the conditions are highly active and metabolic, leading to low pO_2, high pCO_2, high H^+ concentration (lower pH), and higher temperature. These exact conditions are highly favorable for the dissociation of oxygen from oxyhaemoglobin.

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115. (3)

In both lactic acid and alcohol fermentation not much energy is released; less than seven per cent of the energy in glucose is released and not all of it is trapped as high energy bonds of ATP. [Net ATP gain = ATPs synthesized - ATPs utilized = 4 - 2 = 2].

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116. (4)

The principles of photosynthesis as established by Calvin are, at present, being used in studies on renewable resource for energy and materials and basic studies in solar energy research.

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117. (4)

Among the given plants *Orobanche, Striga* and *Cuscuta* are parasitic plants while Venus flytrap and Pitcher plant are insectivorous.

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118. (3)

Ernst Haeckel proposed that vertebrate embryos show vestigial gill slits, supporting evolution.

Karl Ernst von Baer opposed Haeckel, showing embryos do not pass through adult stages of lower animals.

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119. (1)

Let us look at the phenotypic ratio of a Mendelian dihybrid cross (Round/Yellow RrYy×RrYy). The phenotypic ratio in the F_2 generation is:

Round and Yellow: 9/16 (Parental)

Round and Green: 3/16 (Non-parental/Recombinant)

Wrinkled and Yellow: 3/16 (Non-parental/Recombinant)

Wrinkled and Green: 1/16 (Parental)

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120. (4)

Production of pink coloured flowers in *Antirrhinum* sp.: This is an example of incomplete dominance, a pattern of inheritance where the heterozygous phenotype is intermediate between the two homozygous phenotypes. It is a natural genetic phenomenon, not artificial selection.

No production of sugar by allele *i* of gene *I*: This refers to the genetics of the ABO blood group system, which involves multiple alleles. The *i* allele is a recessive, non-functional allele in terms of sugar production, which is a natural biological trait, not a result of human selection.

Pleiotropic effect of a defect in the gene that synthesizes phenylalanine hydroxylase: This refers to Phenylketonuria (PKU). Pleiotropy is a genetic phenomenon where one gene influences multiple, seemingly unrelated phenotypic traits. This is a genetic disorder and not an example of artificial selection.

Production of Sahiwal cows in Punjab: Artificial selection (also known as selective breeding) is the process by which humans breed plants or animals for particular traits. Sahiwal cows are a breed developed specifically for their high milk yield and resistance to tropical heat through human intervention and selective breeding programs, making this the correct answer.

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121. (2)

The classic textbook example of competitive inhibition is the inhibition of succinic dehydrogenase by malonate, which closely resembles the natural substrate succinate in its molecular structure.

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122. (2)

Compositae	-	Basal
Cruciferae	-	Parietal
Malvaceae	-	Axile
Leguminosae	-	Marginal

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123. (3)

All types of RNAs—including messenger RNA (mRNA), ribosomal RNA (rRNA), and transfer RNA (tRNA)—are synthesized from DNA templates through the process of transcription. In prokaryotes, all three types of RNA are synthesized by a single type of DNA-dependent RNA polymerase.

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124. (1)

A person with blood group O can donate to any ABO group, so is called a universal donor.

Group O blood lacks A and B antigens on RBCs, preventing immune reactions in recipients.

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125. (2)

In poriferans, water enters through ostia, passes into spongocoel, and exits via osculum. Ctenophores are marine and diploblastic, with comb plates for locomotion. Platyhelminthes are triploblastic but acoelomate, not pseudocoelomate. Also, they have an incomplete digestive system (no anus). Annelida possess a double ventral nerve cord with paired ganglia. Aschelminthes are generally dioecious, with females often longer than males (sexual dimorphism).

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126. (4)

This is a classic case of natural selection (as seen in Charles Darwin's finches).

After the loss of medium-sized seeds, only large, hard seeds remain.

Finches with larger, stronger beaks will be better able to crack these seeds → higher survival and reproduction.

Over generations, the population will shift toward larger beaks.

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127. (3)

Some organisms breed only once in their lifetime (Pacific salmon fish, bamboo) while others breed many times during their lifetime (most birds and mammals). Some produce a large number of small-sized offspring (Oysters, pelagic fishes) while others produce a small number of large-sized offspring (birds, mammals).

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128. (2)

Biodiversity is the term popularised by the sociobiologist Edward Wilson. A more conservative and scientifically sound estimate made by Robert May places the global species diversity at about 7 million. During his pioneering and extensive explorations in the wilderness of South American jungles, the great German naturalist and geographer Alexander von Humboldt observed that within a region species richness increased with increasing explored area, but only up to a limit. Paul Ehrlich proposed the "rivet popper hypothesis" to explain the importance of biodiversity in ecosystem stability, comparing ecosystem species to rivets on an airplane..

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129. (4)

The nucleolus is the site of ribosomal RNA (rRNA) synthesis and the assembly of ribosomal subunits. A cell with many large nucleoli is producing a massive amount of ribosomes to support active protein synthesis. This structure is found only in eukaryotic cells.

Matches with (4): "eukaryotic cell actively carrying out protein synthesis."

Compartmentalization refers to the presence of membrane-bound organelles (like the nucleus, mitochondria, and ER). Prokaryotic cells lack these membrane-bound organelles, meaning their cytoplasm is not compartmentalized.

Matches with (3): "a prokaryotic cell."

Cell walls are not exclusive to one domain of life. They are found in prokaryotes (e.g., bacteria with peptidoglycan) and many eukaryotes (e.g., plants with cellulose, fungi with chitin, and algae with galactans/mannans).

Matches with (2): "can be a eukaryotic or a prokaryotic cell."

[Contd...

(1) is incorrect because large nucleoli indicate a presence and high activity of ribosome production, not a lack of ribosomes. Also, fungal cells are eukaryotes and are compartmentalized.

(2) is incorrect because large nucleoli indicate that rRNA synthesis is very active, not "compromised."

(3) is incorrect because 40S and 60S ribosomal subunits are characteristic of eukaryotes, which are compartmentalized (contradicting observation ii).

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130. (4)

P700 (in Photosystem I) and P680 (in Photosystem II) are the reaction center chlorophyll *a* molecules. Carotenoids are part of the antenna complex (light-harvesting complex) that captures light energy and transfers it to the reaction center, but they are not part of the reaction center itself. Leucoplasts are colourless plastids primarily involved in the storage of nutrients (such as starch in amyloplasts, fats in elaioplasts, or proteins in aleuroplasts). Carotenoids are typically stored in chromoplasts, which are responsible for the yellow, orange, and red pigments in flowers and fruits. The stress hormone abscisic acid is derived from carotenoids.

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131. (1)

In nephron:

Active reabsorption: Glucose, amino acids, Na^+ (via carriers, ATP-dependent or secondary active transport)

Passive reabsorption: Water (osmosis), urea (diffusion)

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132. (1)

The zona glomerulosa is the outer layer of the adrenal cortex that secretes mineralocorticoids, principally aldosterone, which regulate water and electrolyte balance. Glucocorticoids are mostly from the fasciculata, and androgens from the reticularis. The emergency response is governed by the adrenal medulla.

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133. (3)

Each organized skeletal muscle is made of a number of muscle bundles or fascicles held together by a common collagenous connective tissue layer called the fascia. The sarcolemma lines individual muscle fibres.

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134. (3)

Streptokinase produced by the bacterium *Streptococcus* and modified by genetic engineering is used as a 'clot buster' for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack. Another bioactive molecule, cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus *Trichoderma polysporum*. Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol. Micro-organisms such as *Lactobacillus* and others commonly called lactic acid bacteria (LAB) grow in milk and convert it to curd. During growth, the LAB produce acids that coagulate and partially digest the milk proteins. A small amount of curd added to the fresh milk as inoculum or starter contain millions of LAB, which at suitable temperatures multiply, thus converting milk to curd, which also improves its nutritional quality by increasing vitamin B_{12}.

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135. (4)

In arithmetic growth, following mitotic cell division, only one daughter cell continues to divide while the other differentiates and matures. The simplest expression of arithmetic growth is exemplified by a root elongating at a constant rate. It is characterized specifically by its linear, constant rate of increase (not involving lag phase in growth). Root growth, particularly at the meristematic zone, is generally considered indeterminate because it continues to grow as long as environmental conditions allow and the meristem is active. It can be measured particularly, in terms of length.

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136. (1)

List-I	List-II
EFB	European Federation of Biotechnology
Traditional hybridisation	Inclusion and multiplication of undesirable genes
DNA Polymerase	Synthesizes continuous multiple copies of the gene of interest during PCR.
Lysozyme	Breaks open the bacterial cell wall to facilitate DNA extraction.

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137. (3)

GAA and GAG code for glutamic acid. The genetic code is nearly universal: for example, from bacteria to human UUU would code for Phenylalanine (phe). Some exceptions to this rule have been found in mitochondrial codons, and in some protozoans.

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138. (2)

Hershey and Chase proved that the genetic material (DNA in their case, RNA in this scenario) enters the host cell. The viral genetic material enters the host to direct the synthesis of new viral components. When a phage infects a bacterium, it injects its genetic material (the labelled RNA). This RNA directs the synthesis of new viral RNA and viral proteins using the bacterium's own nucleotides and amino acids. In a standard Hershey-Chase experiment, the labelled DNA (or RNA in this case) is used as a template for replication. The radioactive atoms (like ^{32}P) are part of the original parental viral genome. During the replication of the virus, the original labelled strand remains intact as part of a hybrid molecule or is degraded, but it does not become chemically incorporated into the host bacterium's own chromosomal DNA. The host's genome is a separate entity. Therefore, the radioactivity does not move into the bacterial genome itself. Phosphorus-labelled genetic material enters the bacterial cells (that form the pellet) during the injection process. The supernatant contains the empty protein capsids (labelled with ^{35}S), not the phosphorus-labelled genome. Sulphur labels the protein coat. The protein coat remains outside the bacterial cell and is stripped away during blending. Therefore, ^{35}S is found in the supernatant, not the pellet.

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139. (3)

In *Funaria* (a moss), the sporophyte is partially or fully dependent on the gametophyte for nutrition, not the other way around. The sporophyte is differentiated into a foot, seta, and capsule. The capsule is the specialized structure where meiosis occurs to produce haploid spores. The rhizoids in Funaria are multicellular and branched. The protonema (both primary and secondary) is a stage of the gametophyte, which is haploid (n); therefore, meiosis cannot occur in its cells.

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140. (3)

The SAN generates an action potential that stimulates both the atria to undergo simultaneous contraction (atrial systole). This active pumping phase increases the flow of blood into the ventricles by about 30 percent.

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141. (1)

Cats belong to the order Carnivora, whereas monkeys belong to the order Primata. Both these orders are included in class Mammalia.

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142. (2)

Growth Hormone (GH) promotes the elongation of long bones by acting on the epiphyseal plates (growth plates). In adults, these plates of cartilage calcify and fuse (close) shortly after adolescence due to the influence of sex hormones. Once fused, no further increase in height is possible regardless of GH levels, although excess GH can still cause bone thickening (acromegaly).

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143. (3)

During the electron transport chain of non-cyclic photophosphorylation, plastoquinone (PQ) receives electrons (along with protons from the stroma) and is reduced to plastoquinol (PQH_2). This molecule then diffuses within the thylakoid membrane to transfer electrons to the cytochrome B6f complex and releases protons into the thylakoid lumen. The light reactions (the photochemical phase) of photosynthesis take place in the thylakoid membranes (specifically the grana lamellae and stroma lamellae), not the stroma. The stroma is the site where the Calvin cycle (the biosynthetic/dark reaction phase) of photosynthesis occurs, where the ATP and NADPH produced in the light reactions are used to fix CO_2 into sugars.

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144. (3)

While self-pollination leads to high genetic uniformity, it is not strictly "cloning" (which is asexual reproduction). Self-pollination still involves meiosis and the recombination of alleles from the same parent, allowing for some genetic variation compared to vegetative propagation or asexual reproduction. Since the flowers are closed, external pollinating agents (like bees or wind) cannot access the reproductive parts. Their presence has no effect on pollination in cleistogamous flowers. Because cleistogamy forces continuous self-pollination, the plant is subject to the accumulation of recessive deleterious alleles. Over successive generations, this high rate of inbreeding leads to inbreeding depression, which can reduce the vigor and fertility of the plant population. Cleistogamous flowers are typically bisexual (hermaphrodite) to facilitate self-pollination within the closed flower bud.

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145. (2)

In angiosperms, the seed coat (testa) is derived from the integuments of the ovule. The integuments are maternal tissue (part of the parent plant's sporophyte, 2n. Since the seed coat is formed entirely from the maternal parent's tissues before fertilization and development of the zygote, its phenotype (including colour) is determined solely by the genotype of the female parent. The genes from the pollen (male parent) do not influence the maternal tissue that forms the seed coat. Though the parents contain two alleles during gamete formation, the factors or alleles of a pair segregate from each other such that a gamete receives only one of the two factors.

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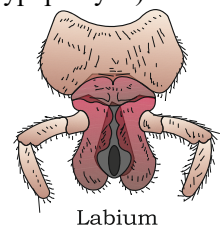
146. (2)

Chondrichthyes (cartilaginous fishes) uniquely have a ventrally located mouth, tough skin with placoid scales, and no air bladder (forcing them to swim constantly). Osteichthyes have a terminal mouth, operculum, cycloid/ctenoid scales, and an air bladder.

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147. (3)

The mouthparts consisting of a labrum (upper lip), a pair of mandibles, a pair of maxillae and a labium (lower lip). A median flexible lobe, acting as tongue (hypopharynx).



Labium

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148. (3)

The limbic lobe or limbic system (including the amygdala and hippocampus), along with the hypothalamus, is involved in the regulation of sexual behaviour, expression of emotional reactions, and motivation.

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149. (2)

Katherine Esau reported in her early publications that the curly top virus spreads through a plant via the food-conducting or phloem tissue. The pathogens for rest of the given diseases are not dependent on phloem for spread within the plant.

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150. (3)

In water, skin acts as an aquatic respiratory organ (cutaneous respiration). Dissolved oxygen in the water is exchanged through the skin by diffusion.

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151. (3)

Monocot stems possess numerous vascular bundles scattered in the ground tissue. Monocot stems have a well-developed ground tissue system consisting of parenchyma cells that fill the spaces between vascular bundles. Monocots possess complex tissues, specifically xylem and phloem, which are the components of their vascular bundles. Since monocots do not undergo secondary growth, they do not develop a cork cambium.

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152. (3)

A reporter enzyme (such as β-galactosidase or GFP) is used to visually or quantitatively detect gene expression. When foreign DNA is successfully taken up and expressed, the reporter gene produces a measurable signal (colour/fluorescence), helping distinguish transformed cells from non-transformed ones, in addition to selectable markers like antibiotic resistance genes.

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153. (4)

Sporozoans: This includes diverse organisms that have an infectious spore-like stage in their life cycle. The most notorious is *Plasmodium* (malarial parasite) which causes malaria. Amoeboid protozoans: Some of them such as *Entamoeba* are parasites.

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154. (4)

Members of phycomycetes are found in aquatic habitats and on decaying wood in moist and damp places or as obligate parasites on plants. The mycelium is aseptate and coenocytic.

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155. (2)

Decomposers include bacteria (prokaryotes, no nuclear membrane) and fungi (eukaryotes, membrane-bound nucleus). Detritivores (e.g., earthworm) break down detritus into smaller particles. Bacterial and fungal enzymes degrade detritus into simpler inorganic substances. This process is called as catabolism. Warm and moist environment favour decomposition whereas low temperature and anaerobiosis inhibit decomposition resulting in build up of organic materials.

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156. (3)

When migration of a section of a population to another place occurs, gene frequencies change. If this gene migration happens multiple times, it is specifically called gene flow.

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157. (4)

MALT actually constitutes about 50 percent of the lymphoid tissue in the human body, not 20 percent.
Bone marrow is the primary hematopoietic organ producing RBCs, WBCs, and platelets.
Thymus is bilobed and undergoes involution after puberty.
Spleen is a secondary lymphoid organ and also stores RBCs and platelets.

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158. (1)

List-I	List-II
Oogonia	Gamete mother cells formed during embryonic stage.
Primary Oocyte	Cells temporarily arrested in the prophase-I stage of meiotic division.
Secondary Oocyte	A large haploid cell formed by the unequal first meiotic division.
Graafian Follicle	The fully mature tertiary follicle that ruptures during ovulation.

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159. (2)

List-I	List-II
Multipolar neuron	Cerebral cortex
Bipolar neuron	Retina of the eye
Unipolar neuron	Usually in the embryonic stage
Myelinated nerve fibre	Spinal and cranial nerves

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160. (3)

The main challenge for producing insulin using rDNA techniques was getting the insulin assembled into a mature form (without the C peptide). Eli Lilly produced chains A and B separately, extracted them, and combined them by creating disulfide bonds to form mature human insulin.

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161. (2)

During double fertilization, one male gamete fuses with the two polar nuclei (or the secondary nucleus) in the central cell to form the Primary Endosperm Nucleus (PEN). This process is called triple fusion. The central cell then develops into the endosperm, which serves as a nutritive tissue for the developing embryo. The seed coat is derived from the integuments of the ovule. The antipodal cells usually degenerate after fertilization. The zygote (formed by the fusion of one male gamete and the egg cell) is diploid (2n) and develops into the embryo.

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162. (2)

In a polysaccharide chain like glycogen, the right end is strictly called the reducing end, and the left end is called the non-reducing end. (N and C terminals refer to proteins).

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163. (3)

The second heart sound (dub) is directly associated with the closure of the semilunar valves, which occurs when the ventricles relax (ventricular diastole) and the ventricular pressure falls, preventing the backflow of blood from the aorta and pulmonary artery.

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164. (4)

The nodules on the roots of leguminous plants are formed by the symbiotic association of *Rhizobium*. These bacteria fix atmospheric nitrogen into organic forms, which is used by the plant as nutrient. Other bacteria can fix atmospheric nitrogen while free-living in the soil (examples *Azospirillum* and *Azotobacter*), thus enriching the nitrogen content of the soil. Fungi are also known to form symbiotic associations with plants (mycorrhiza). Many members of the genus *Glomus* form mycorrhiza. Lichens are symbiotic associations i.e. mutually useful associations, between algae and fungi.

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165. (3)

Polysiphonia: A genus of red algae (Rhodophyceae). Being an alga, it is the most primitive among the options with thalloid body organisation, lacking true roots, stems, and leaves.

Polytrichum: A genus of mosses (Bryophyta). These are non-vascular land plants. While more advanced than algae, they lack vascular tissues (xylem and phloem).

Salvinia: A genus of pteridophytes (ferns). Pteridophytes are vascular plants, which makes them more advanced than bryophytes and algae. However, they are seedless plants.

Wolffia: A genus of minute aquatic flowering plants (Angiosperms). Angiosperms (flowering plants) are the most highly evolved and advanced group in the plant kingdom because they produce seeds, fruits, and have highly specialized vascular systems.

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166. (3)

List-I	List-II
Many bony fishes	Ammonotelic (excrete ammonia mainly by diffusion)
Mammals	Ureotelic (excrete urea processed by kidneys)
Birds	Uricotelic (excrete uric acid paste)
Glomerulus	A tuft of capillaries formed by the afferent arteriole

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167. (2)

An antibody has two small light chains and two longer heavy chains (H_2L_2). They are found in the blood (humor), it is called the humoral immune response, not the cell-mediated immune response (which is governed by T-cells).

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168. (2)

Living cells of a complex plant tissue: Complex tissues are composed of more than one type of cell. Xylem parenchyma are the only living cells found within the xylem (a complex tissue).

Dead cells of a complex plant tissue: Bast fibres (phloem fibres) are sclerenchymatous cells found in the phloem (a complex tissue) and are dead at maturity.

Living cells of simple plant tissue: Simple tissues consist of only one type of cell. Collenchyma (found in the hypodermis of dicot stems) is a living, simple mechanical tissue.

Dead cells of simple plant tissue: Sclerenchyma is a dead simple tissue. The hypodermis of a monocot stem is characteristically composed of sclerenchyma.

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169. (2)

Thyroid follicles synthesize two hormones: thyroxine (T4) and triiodothyronine (T3) (actually they are iodinated amino acid-derived hormones, called amine hormones.

Diabetes insipidus is caused by ADH deficiency or kidney insensitivity to ADH, not insulin.

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170. (4)

Muscle contraction is regulated by the troponin–tropomyosin complex.

At rest: Tropomyosin blocks myosin-binding sites on actin.

During contraction: Ca^{2+} ions bind to troponin, causing a conformational change. This shifts tropomyosin → exposes binding sites → actin–myosin interaction occurs.

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171. (1)

The basal body of cilia or flagella has a "9+0" arrangement of microtubules. It consists of 9 triplets of microtubules arranged in a circle = 27 microtubules. The axoneme of a eukaryotic flagellum has a "9+2" arrangement. This consists of 9 peripheral doublets (9×2=18) and 2 central singlets (2×1=2). Thus, 18+2=20 microtubules. Eukaryotic ribosomes are of the 80S type, consisting of 60S and 40S subunits. Prokaryotic ribosomes are of the 70S type, consisting of 50S and 30S subunits.

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172. (4)

For a gene with two alleles:
Frequency of dominant allele = p
Frequency of recessive allele = q
$p + q = 1$
Genotype frequencies:
P^2 (homozygous dominant)
2pq (heterozygous)
Given:
Total population = 1000
White flowers = recessive phenotype = q^2 = 490/1000 = 0.49
Step 1: Find q $q^2 = 0.49 \rightarrow q = 0.7$
Step 2: Find p $p = 1 - q = 1 - 0.7 = 0.3$
Step 3: Find heterozygous frequency (2pq) $2pq = 2 \times 0.3 \times 0.7 = 0.42$
Step 4: Number of heterozygous plants $0.42 \times 1000 = 420$

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173. (2)

In the mitochondrial Electron Transport System (ETS), Complex I (NADH dehydrogenase), Complex III (Cytochrome bc_1 complex), and Complex IV (Cytochrome *c* oxidase) are proton pumps that move protons from the mitochondrial matrix into the intermembrane space. Complex II (Succinate dehydrogenase), however, does not contribute to the proton gradient directly. Complex II is indeed a membrane-bound enzyme that contains Fe-S (Iron-Sulfur) centers and FAD. These components are essential for the transfer of electrons from succinate to ubiquinone.

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174. (2)

The first step of decomposition of detritus is fragmentation. The last step of decomposition of detritus is mineralisation.

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175. (2)

Predators also help in maintaining species diversity in a community, by reducing the intensity of competition among competing prey species. Besides acting as 'conduits' for energy transfer across trophic levels, predators play other important roles. They keep prey populations under control. But for predators, prey species could achieve very high population densities and cause ecosystem instability.

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176. (3)

Ecosystem varies greatly in size from a small pond to a large forest or a sea. Nutrient cycles, food chains and webs are created as a result of the energy flows within the ecosystem. Species composition and stratification are the two main structural features of an ecosystem. Productivity, decomposition, energy flow, and nutrient cycling are the four important components of an ecosystem. Saprophytes are not given any place in ecological pyramids even though they play a vital role in the ecosystem. The ecosystems are not exempt from the First and Second Law of thermodynamics.

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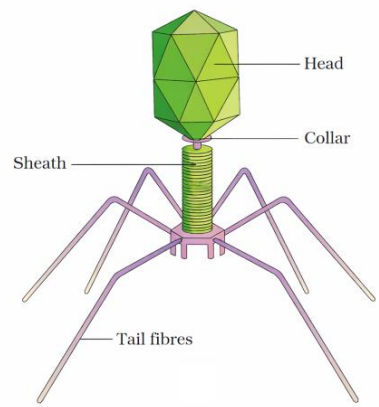
177. (1)

In monocot (isobilateral) leaves, the guard cells are dumbbell-shaped, not bean-shaped (which are characteristic of dicots). In an isobilateral leaf, the mesophyll is not differentiated into palisade and spongy parenchyma; instead, it is composed of uniform mesophyll cells.

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178. (4)

The diagram in the given question is that of a bacteriophage. In the five kingdom classification of Whittaker there is no mention of lichens and some acellular organisms like viruses, viroids and prions. Bacteriophages are viruses that infect the bacteria. Viruses did not find a place in classification since they are not considered truly 'living', if we understand living as those organisms that have a cell structure. The viruses are non-cellular organisms that are characterised by having an inert crystalline structure outside the living cell. Once they infect a cell, they take over the machinery of the host cell to replicate themselves, killing the host. The genome of *E. coli* has 4.6×10^6 bp. virus is a nucleoprotein and the genetic material is infectious. In case of the shown bacteriophage, genetic material is present in its head (A).



Bacteriophage

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179. (4)

In pea and bean flowers, there are five petals, the largest (standard) overlaps the two lateral petals (wings) which in turn overlap the two smallest anterior petals (keel); this type of aestivation is known as vexillary or papilionaceous. In plants such as bean, gram and pea, the endosperm is not present in mature seeds and such seeds are called non-endospermous. The flower can be divided into two similar halves only in one particular vertical plane i.e. zygomorphic. Bean is a member of family Leguminosae and is characterised by diadelphous (1+9 arrangement), where nine stamens are fused into a bundle and one stamen remains free. While the leaves are indeed alternate (phyllotaxy), they exhibit reticulate venation (a characteristic of dicots), meaning the veins are not all the same size; they form a complex network with a prominent midrib and smaller veinlets.

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180. (2)

Cnidarians like *Obelia* exist in both forms and exhibit alternation of generations (Metagenesis), i.e., polyps produce medusae asexually and medusae form the polyps sexually.

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